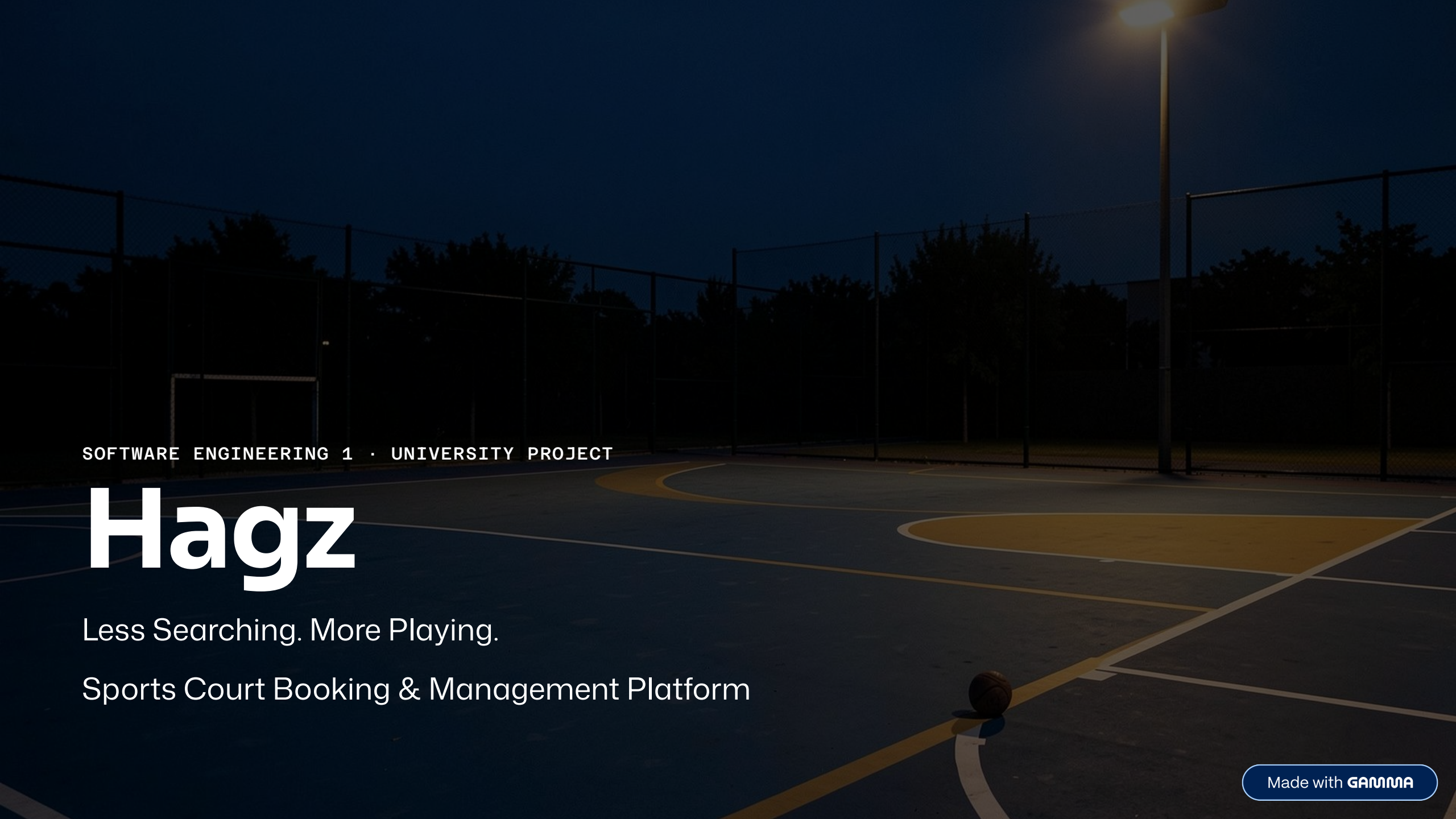




SOFTWARE ENGINEERING 1 · UNIVERSITY PROJECT

Hagz

Less Searching. More Playing.

Sports Court Booking & Management Platform

From Searching to Booking

THE FRICTION

01 Search

02 Contact

03 Ask about availability

04 Wait

05 Book

THE HAGZ FLOW

01 Find

02 Explore

03 Check availability

04 Select time

05 Book

Hagz makes finding and booking sports courts simpler, clearer, and more organized.

Who Uses Hagz?

PLAYER

Find a court. Book with clarity.

- Search courts
- Filter by area and price
- View details and availability
- Book, cancel, and view bookings

COURT OWNER

Manage the court lifecycle.

- Add and manage courts
- Upload photos
- Set prices and availability
- Accept or reject booking requests

ADMINISTRATOR

Keep the marketplace trusted.

- Review submitted courts
- Approve or reject courts
- Manage users

Requirements & Project Scope

FUNCTIONAL REQUIREMENTS

- **Authentication:** registration, login, role-based authorization
- **Court management:** courts, photos, pricing, availability
- **Booking:** view slots, create, cancel, accept, or reject bookings
- **Administration:** review courts, approve or reject courts, manage users

CURRENT V1

7

Core capabilities

Authentication · RBAC · Court Management · Search & Filtering · Availability · Booking · Admin Moderation

Use Case Analysis

Three actors interact with grouped Hagz system boundaries.

HAGZ SYSTEM

Player

Register · Login · Search
Courts · Filter Courts · View
Court Details · View
Availability · Create Booking ·
Cancel Booking · View
Bookings

Court Owner

Register · Login · Add Court ·
Manage Court · Define
Availability · View Booking
Requests · Accept Booking ·
Reject Booking

Administrator

Login · Review Court ·
Approve Court · Reject Court ·
Manage Users

System Architecture

A simple three-layer architecture with file storage for court images.

Layer	Components
Client layer	Player Web App · Court Owner Web App · Admin Dashboard
Backend layer · REST API	Authentication · Authorization / RBAC · User Management · Court Management · Availability Management · Booking Management · Validation · Business Logic
Data layer	Database
File storage	Court Images

Core System Design

SIMPLIFIED ERD

Entity	Fields
User	id · name · email · passwordHash · role · status
Court	id · ownerId · name · address · area · price · status
CourtImage	id · courtId · imageUrl
Availability	id · courtId · day · startTime · endTime
Booking	id · playerId · courtId · date · startTime · endTime · status

BOOKING WORKFLOW

1. Player searches for a court
2. Selects a court and checks availability
3. Selects date and time
4. Creates a booking
5. Court owner accepts or rejects
6. Status moves from Pending to Accepted or Rejected
7. Eligible bookings can be cancelled by the player

PRODUCT EXPERIENCE

UI/UX Design

Hagz is a real digital product: minimal, clear, and organized around the booking flow.

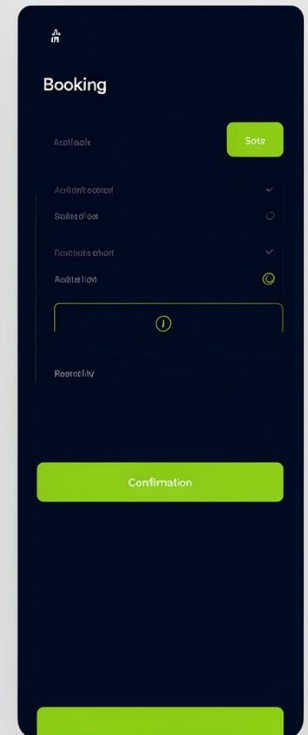
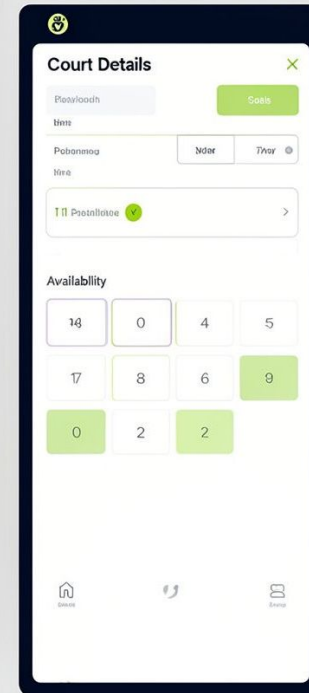
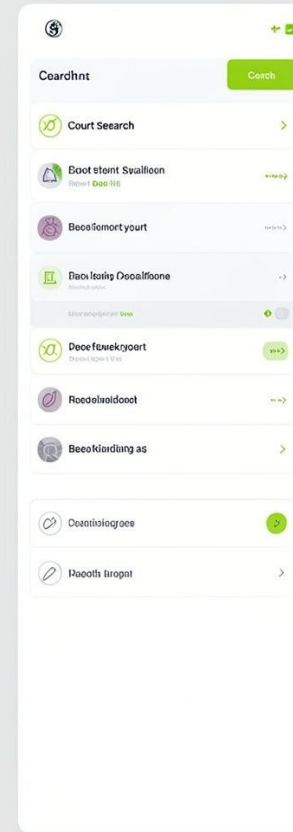
PLAYER FLOW

Find Court
Court Details
Booking
Confirmation
My Bookings

OPERATIONS FLOW

Owner Dashboard
Booking Requests
Admin Dashboard
Pending Courts

MINIMAL PREMIUM INTERFACE · DEEP NAVY + LIME
IDENTITY · CONSISTENT COMPONENTS · STRONG VISUAL
HIERARCHY · RESPONSIVE WEB EXPERIENCE



Technology & Quality

TECHNOLOGY STACK

- **Frontend:** React · TypeScript / JavaScript · Tailwind CSS
- **Backend:** Node.js · Express.js · REST API
- **Database:** MongoDB · Mongoose
- **Development:** Git · GitHub · Postman · Docker

QUALITY ATTRIBUTES

Security	Performance
Usability	Reliability
Maintainability	Scalability
Availability	Compatibility

Quality foundation: Authentication + Authorization + RBAC + Input Validation + Password Hashing

What's Next?

Hagz transforms sports court booking from a fragmented process into a simple digital experience.



FUTURE SCOPE · SEPARATE FROM CURRENT V1

Online Payments	Recommendation Systems
Maps & Navigation	Mobile Application
Notifications	Analytics
Advanced Search	Loyalty Features

**Less
Searching.
More Playing.**

Hagz